

IEM KOLKATA
Linear Algebra
Matrices & Determinants

Q1) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then $\det(A^{-1})$ is

- (a) 0.25 (b) 0.50 (c) 0.75 (d) 1.25

Q2) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$.

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?

- (a) $\det(A) = \det(B)$
 (b) $\det(B) = -\det(A)$
 (c) $\det(A) = 0$
 (d) $\det(AB) = \det(A) + \det(B)$

Q3) The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has $\det(A) = 100$ and $\text{trace}(A) =$

14. The value of $|a - b|$ is

- (a) 1 (b) 2 (c) 3 (d) 4

Q4) If any two columns of a determinant $= \begin{vmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{vmatrix}$ are interchanged,

which one of the following statements regarding the value of the determinant is CORRECT?

- (A) Absolute value remains unchanged but sign will change
- (B) Both absolute value and sign will change
- (C) Absolute value will change but sign will not change
- (D) Both absolute value and sign will remain unchanged

Q5) Consider the following two statements with respect to the matrices $A^{m \times n}, B^{n \times m}, C^{n \times n}, D^{n \times n}$.

Statement 1 : $\text{tr}(AB) = \text{tr}(BA)$

Statement 2 : $\text{tr}(CD) = \text{tr}(DC)$

where $\text{tr}(\)$ represents the trace of a matrix. Which one of the following holds?

- (A) Statement 1 is correct and Statement 2 is wrong
- (B) Statement 1 is wrong and Statement 2 is correct
- (C) Both Statement 1 and Statement 2 are correct
- (D) Both Statement 1 and Statement 2 are wrong

Q6) Let $ABC = I$, then $B^{-1} = ?$

Q7) Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then $B^{-1} = ?$

Q8) X, Y are singular matrices $\ni XY = Y$ and $YX = X$. Then show that $X^2 + Y^2 = X + Y$.

Q9) If $|A^{3 \times 3}| = 3$, then $|(2A)^{-1}| = ?$

Q10) If $|A^{3 \times 3}| = 4$, then $\text{adj}(\text{adj}(A)) = ?$

Q11) Consider the system of simultaneous equations:

$$x + 2y + z = 6$$

$$2x + y + 2z = 6$$

$$x + y + z = 5$$

The system has:

- (a) unique solution
- (b) infinite no of solutions
- (c) no solution
- (d) exactly two solutions

Q12) Consider the system of simultaneous equations:

$$\begin{bmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ 5 \\ 7 \end{bmatrix}$$

Notice that the second and the third columns of the coefficient matrix are linearly dependent. For how many values of α , does this system of equations have infinitely many solutions?

- (a) 0 (b) 1 (c) 2 (d) infinitely many

Q13) Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3 \times 3}$ and $Q \in \mathbb{R}^{3 \times 1}$.

Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Which of the following statement(s) is/are TRUE?

- (a) The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
- (b) If P is invertible, then both L and U are invertible.
- (c) If P is singular, then at least one of the diagonal elements of U is zero.
- (d) If P is symmetric, then both L and U are symmetric.